Rainfall Shocks, Household Hardship, and Credit: Evidence from Eight Developing Countries*

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October 29, 2026

Abstract

Weather shocks strike entire communities at once, limiting informal insurance precisely when it is most needed. We study how households cope with rainfall shocks during a decade of expanding financial access, linking 370,242 geocoded adults in eight countries to satellite rainfall. A one-standard-deviation rainfall shortfall raises the probability that a child is sent home from school for unpaid fees by 19 percent and economic vulnerability by 8 percent. Effects are proportionally similar for farm and non-farm households, consistent with a shock to the local economy. Households respond by borrowing, especially from microfinance institutions in South Asia, while bank accounts, savings, insurance, and transfers from relatives do not respond. Credit is the margin of adjustment, and it does not prevent hardship.

JEL codes: D14, G21, O12, O16, Q54

Keywords: rainfall shocks, risk coping, household hardship, microfinance, financial inclusion, credit

*We thank Abu Shonchoy for providing access to the Financial Inclusion Insights data, and the Climate Hazards Center at the University of California, Santa Barbara, for making the CHIRPS rainfall data publicly available. All errors are our own.

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I. Introduction

Poor households in developing countries face large and frequent income shocks, and weather is among the most consequential. A failed rainy season reduces harvests, depresses rural wages, and raises food prices (Jayachandran, 2006; Dell, Jones, and Olken, 2014). Unlike illness or job loss, rainfall shocks are spatially correlated: they strike neighbors at the same time. This is precisely the circumstance in which informal insurance — transfers among relatives, friends, and neighbors — is least able to help (Townsend, 1994; Udry, 1994; Fafchamps and Lund, 2003). How households cope with such shocks, and at what cost, is a central question in development economics (Morduch, 1995; Dercon, 2002).

Over the past decade, governments, donors, and the private sector have invested heavily in one answer: financial inclusion. Between 2014 and 2018, India opened hundreds of millions of bank accounts, mobile money spread across East Africa, and microfinance continued to expand in South Asia (Demirgüç-Kunt et al., 2018; Suri and Jack, 2016). A central promise of this expansion is resilience — that accounts, savings, insurance, and cheap transfers would allow households to absorb shocks without sacrificing consumption or their children's schooling. Whether this promise is visible in household behavior, outside a small number of well-studied settings (Jack and Suri, 2014; Riley, 2018; Abiona and Koppensteiner, 2022), remains an open question.

We study how households in eight countries — Bangladesh, India, Indonesia, Kenya, Nigeria, Pakistan, Tanzania, and Uganda — respond to rainfall shocks, using nationally representative survey data on 370,242 adults from the Financial Inclusion Insights (FII) program for 2014–2018. The surveys record hardship, borrowing, and the full range of financial instruments for the same respondents, and include GPS coordinates for each survey cluster. We link them to high-resolution satellite rainfall and measure, for each cluster, rainfall in the 12 months before fieldwork relative to that location's own history for the same months since 1981.

To organize the analysis, we describe a simple framework in which a household facing a rainfall shortfall can absorb the shock through savings, insurance, transfers, or credit, or else bear it as hardship. The framework yields four predictions. First, shortfalls raise hardship when coping is incomplete. Second, if rainfall affects the local economy through wages and prices rather than only farm output, hardship rises among non-farm as well as farm households. Third, because the shock is common to a community, transfers from relatives are unlikely to rise, and households turn to the instruments that remain available. Fourth, the instruments that respond reveal which margins of coping are actually used.

We estimate the effects of rainfall using fixed effects for 0.5° grid cells, which absorb permanent differences across places, and for $2° \times 2°$ regions interacted with survey year, which absorb national and regional shocks and trends. Identification comes from local, year-to-year deviations in rainfall among nearby locations surveyed in the same year. We find that a one-standard-deviation rainfall shortfall increases the probability that a household had a child sent home from school for unpaid fees by 1.4 percentage points, or 19 percent of the mean. It raises the probability of being economically vulnerable — experiencing any of eight shortages in the past six months — by 2.7 points (8 percent), and of going without food or cooking fuel by 13 percent each.

These effects are not confined to farmers. Measured relative to baseline levels, hardship rises by similar proportions among farm and non-farm households: the share of households with a child sent home for unpaid fees increases by 16 percent among households involved in agriculture and 18 percent among those that are not. This is the pattern expected if rainfall acts as a shock to the local economy — reducing labor demand, wages, and local spending while raising food prices — rather than only to farm income.

Households respond to these shocks by borrowing. A one-standard-deviation shortfall raises the probability of having borrowed in the past year by 2.2 points (6 percent) and of borrowing from non-bank financial institutions (NBFIs), mostly microfinance institutions, by 1.0 point (16 percent). The NBFI response, the only borrowing source that remains

significant with state-by-year fixed effects, is concentrated in Bangladesh and India, where microfinance is most widespread. Microfinance loans are designed to finance enterprises; their use after poor rainfall suggests that households draw on the most accessible formal credit to meet consumption needs.

By contrast, the instruments emphasized by financial inclusion policy do not respond. Bank account ownership and use, formal saving, insurance, and transfers received from relatives and friends are unaffected by rainfall. These nulls are precisely estimated: using spatially robust standard errors and two one-sided tests, we can rule out effects of a shortfall larger than 1.5 percent of the mean on bank account ownership and increases larger than 3.8 percent in formal saving. Credit is the margin on which households adjust. Since hardship rises despite this borrowing, credit is not sufficient to protect them.

The results survive a battery of identification and robustness checks. Rainfall in the year after fieldwork, which cannot have affected survey responses, has no effect on any outcome. Respondent characteristics are balanced with respect to rainfall, which argues against changes in who is surveyed. The estimates hold with state-by-year fixed effects, which absorb state-level drought relief and loan waivers; with finer 1° region-by-year effects; with rainfall measured relative to local trends; with Conley standard errors allowing spatial correlation up to 500 kilometers; and, for school fees, economic vulnerability, and borrowing, when excluding India, which accounts for 62 percent of the sample.

In magnitudes, the effects are economically meaningful. For every 100 households, a one-standard-deviation shortfall implies roughly one to two additional households with a child sent home from school for unpaid fees and nearly three additional economically vulnerable households, alongside two additional borrowers. Because hardship reaches non-farm households and formal instruments do not respond, these findings suggest that access alone has not equipped households to absorb covariate shocks, and that policies targeted only at farmers will miss much of the hardship that rainfall shocks cause.

This paper contributes to three literatures. First, we add to a long literature on how

households cope with income risk (Paxson, 1992; Rosenzweig and Wolpin, 1993; Townsend, 1994; Udry, 1994; Morduch, 1995; Kochar, 1999; Dercon, 2002). Much of this literature studies a single country or a small number of villages, and typically observes either welfare outcomes or coping strategies, but not both. We provide comparable estimates across eight countries on two continents and document, in the same data, the welfare cost of shocks, the margin on which households adjust, and the margins on which they do not. Our finding that hardship rises proportionally among non-farm households extends the evidence that weather shocks propagate through rural labor and goods markets (Jayachandran, 2006).

Second, we contribute to the literature on financial inclusion (Burgess and Pande, 2005; Dupas and Robinson, 2013; Banerjee et al., 2015; Suri and Jack, 2016). Experimental evaluations have found modest average effects of expanded access to credit and accounts. We complement this evidence by studying the moment when financial tools should matter most — a covariate shock — and show that accounts, savings, and insurance remain unchanged while microfinance credit responds. To our knowledge, this is the first multi-country evidence linking weather shocks to the full portfolio of household financial behavior.

Third, we contribute to the literature on mobile money and risk sharing (Jack and Suri, 2014; Blumenstock, Eagle, and Fafchamps, 2016; Riley, 2018; Abiona and Koppensteiner, 2022; Lee et al., 2021). Influential studies in Kenya and Tanzania show that mobile money facilitates transfers that help households smooth consumption after shocks. Across our eight countries, transfers from relatives do not rise after rainfall shortfalls, suggesting that remittance-based insurance does not generalize readily beyond settings where mobile money is near-universal.

The remainder of the paper proceeds as follows. Section II describes the setting, and Section III the data. Section IV presents the conceptual framework, and Section V the empirical strategy. Section VI reports the main results and robustness checks, Section VII examines mechanisms and alternative explanations, Section VIII discusses magnitudes and policy implications, and Section IX concludes.

II. Background

A. Rainfall Risk in Low-Income Economies

Agriculture remains a major source of employment and income in all eight countries we study, and in most of them a large share of cultivation is rain-fed; even where irrigation is widespread, rainfall governs water availability and the timing of the agricultural season. Rainfall therefore shapes household incomes directly, through harvests, and indirectly, through the local economy. When rains fail, agricultural labor demand and rural wages fall, the incomes of traders and service providers who depend on local spending decline, and food prices rise (Jayachandran, 2006; Dell, Jones, and Olken, 2014). Because these effects operate through local markets, they reach households regardless of whether they farm.

Covariate shocks of this kind are difficult to insure informally. Risk-sharing arrangements among relatives and neighbors work well against idiosyncratic shocks, such as illness, but break down when all members of a network are hit at once (Townsend, 1994; Udry, 1994; Fafchamps and Lund, 2003). Households facing a common shock must instead rely on their own savings, on formal insurance, on transfers from outside the affected area, or on credit — or reduce consumption. One of the most visible forms of reduced consumption in low-income countries is the inability to pay school fees: even where primary schooling is nominally free, levies, examination fees, and secondary-school fees are common, and children are sent home when they go unpaid.

B. The Expansion of Financial Access

The years 2014–2018 saw a rapid but uneven expansion of financial access. India's Pradhan Mantri Jan Dhan Yojana, launched in 2014, opened bank accounts for a large share of previously unbanked adults. In East Africa, mobile money, pioneered by Kenya's M-Pesa, became the dominant financial service for many households (Suri and Jack, 2016). In Bangladesh and India, microfinance institutions had long provided small loans, primarily to women and nominally for enterprise investment (Banerjee et al., 2015). Figure 1 shows account own-

ership by country and wave in our data. Bank account ownership rose sharply in India and Indonesia, and mobile money grew in Kenya, Tanzania, Uganda, and Bangladesh. This expansion is the backdrop against which we ask how households cope with rainfall shocks.

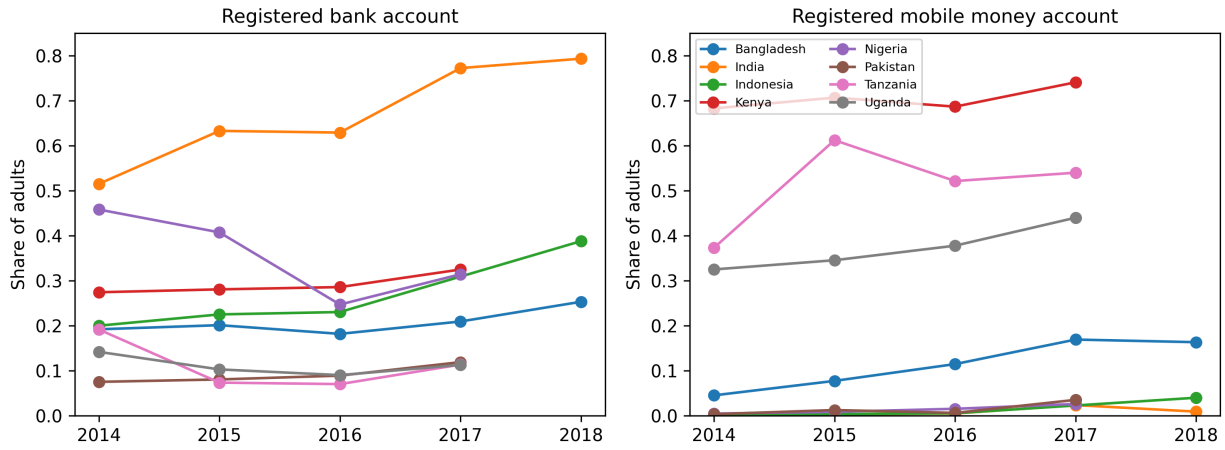


Figure 1. Account Ownership by Country and Survey Wave

Notes: Unweighted shares of adults with a bank account or mobile money account registered in their name. *Source:* FII surveys.

III. Data and Summary Statistics

A. Household Survey Data

The FII program conducted nationally representative surveys of adults aged 15 and older in eight countries, implemented by InterMedia (later Kantar) with support from the Bill & Melinda Gates Foundation.[1] Each wave draws an independent sample using a stratified multistage design based on national census frames, with one adult per household selected by Kish grid. The surveys are repeated cross-sections and include anonymized GPS coordinates at the household-cluster level. Poverty status is measured using the Poverty Probability Index.[2]

Our hardship outcomes record whether, in the past six months, the household had a child sent home from school because of unpaid fees, went without food, or went without

[1] FII dataset documentation: https://fii-codebook.drivendata.org/.

[2] PPI documentation and country-specific design memos: https://www.povertyindex.org/get-started-ppi.

cooking fuel; these were asked in 2015 and 2016. Economic vulnerability, an indicator for experiencing any of eight such shortages, is observed in 2014–2016. Our borrowing outcomes, observed in all waves, are indicators for having borrowed in the past year and for borrowing from an NBFI (a microfinance institution, cooperative, savings and credit cooperative, or post office bank). We also observe bank and mobile money accounts, formal and informal saving, insurance, and whether the respondent received money from relatives or friends in the past year.

Our sample consists of all respondents in waves from 2014 to 2018 with valid coordinates and non-missing controls: 370,242 adults in 20,625 survey clusters, 2,159 grid cells, and 34 country-waves.[3]

B. Rainfall Data

We measure rainfall using the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS), version 2.0 (Funk et al., 2015), which combines satellite imagery with station observations at 0.05° resolution from 1981 onward. We retrieve monthly totals at every second pixel (an effective resolution of 0.1°, about 11 km) and assign each survey cluster to the nearest pixel with valid data. For each cluster and wave, we compute rainfall in the 365 days before the start of fieldwork, using start dates from InterMedia's reports for 28 of 34 country-waves and an imputed September start for the remaining six (Appendix Table A.3). Our shock is the z-score of this total relative to the same 12 calendar months in each year from 1981 to 2012 at the same pixel, so that it measures how unusual rainfall was relative to what households in that location expect. A 12-month window spans a full agricultural cycle.

[3]The 2013 waves lack the employment variable used as a control and, for Nigeria and Tanzania, GPS coordinates, and are excluded. The country columns of Table 1 sum to 2,199 grid cells because 40 cells straddle national borders.

C. Summary Statistics

Table 1 reports summary statistics. India accounts for 62 percent of respondents because of its large sample design. Hardship is common: on average, 7.4 percent of households had a child sent home for unpaid fees in the past six months, and 34.7 percent were economically vulnerable. Thirty-seven percent of adults borrowed in the past year, and 6.3 percent borrowed from an NBFI. Account ownership varies widely across countries, from 9 percent in Pakistan to 67 percent in India, and mobile money use is concentrated in Kenya, Tanzania, and Uganda.

Table 1. Summary Statistics

	All	Bangladesh	India	Indonesia	Kenya	Nigeria	Pakistan	Tanzania	Uganda
Demographics									
Age (years)	37	37	38	41	35	34	36	36	35
Female	0.544	0.540	0.547	0.600	0.604	0.456	0.480	0.564	0.614
Rural	0.666	0.694	0.685	0.478	0.618	0.625	0.665	0.688	0.731
Married	0.709	0.808	0.742	0.496	0.613	0.554	0.779	0.622	0.609
Works	0.510	0.408	0.482	0.556	0.632	0.623	0.458	0.751	0.700
Borrowing									
Any borrowing, past year	0.370	0.405	0.365	0.518	0.561	0.267	0.172	0.317	0.525
Borrowed from a non-bank fin. institution	0.063	0.180	0.054	0.086	0.099	0.027	0.016	0.043	0.065
Hardship, past six months									
Child sent home for unpaid fees	0.074	0.018	0.042	0.013	0.411	0.119	0.021	0.120	0.424
Economic vulnerability (any shortage)	0.347	0.377	0.285	0.234	0.733	0.469	0.236	0.566	0.808
Went without food	0.161	0.180	0.117	0.082	0.359	0.307	0.041	0.392	0.456
Went without cooking fuel	0.140	0.137	0.118	0.075	0.307	0.233	0.036	0.267	0.324
Accounts, savings, insurance									
Has bank account	0.499	0.207	0.671	0.264	0.292	0.355	0.091	0.112	0.112
Active mobile money user	0.059	0.094	0.007	0.010	0.642	0.009	0.012	0.415	0.320
Saves formally	0.347	0.253	0.395	0.308	0.518	0.350	0.054	0.254	0.247
Saves informally	0.100	0.044	0.074	0.116	0.288	0.174	0.141	0.078	0.289
Has insurance	0.140	0.072	0.153	0.271	0.266	0.052	0.030	0.171	0.030
Rainfall									
Rainfall, 12 months before survey (mm)	1368	2380	1260	2397	1079	1334	455	1080	1271
Long-run mean, same window (mm)	1321	2301	1196	2406	1098	1428	371	999	1250
Rainfall z-score	0.223	0.220	0.249	0.004	-0.098	-0.468	1.006	0.395	0.162
Drought ($z < -1$)	0.094	0.040	0.076	0.157	0.191	0.271	0.022	0.118	0.089
Wet ($z > 1$)	0.203	0.187	0.200	0.134	0.121	0.076	0.470	0.299	0.155
Respondents	370,242	30,000	230,822	24,815	12,118	24,395	24,000	12,090	12,002
Survey clusters	20,625	3,358	6,474	2,424	1,217	2,357	2,398	1,206	1,191
Grid cells	2,159	69	989	264	133	254	180	227	83
Survey waves	34	5	5	4	4	4	4	4	4

Notes: Unweighted means. The sample covers FII survey waves 2014–2018 with GPS coordinates and non-missing controls. Economic vulnerability is observed in 2014–2016; the other hardship items in 2015–2016. Rainfall is CHIRPS v2.0 total precipitation in the 12 months before the start of fieldwork at each respondent's 0.1° pixel; the z-score standardizes it against the same 12-month window in 1981–2012 at the same pixel. Grid cells are $0.5° \times 0.5°$ (PRIO-GRID resolution).

IV. Conceptual Framework

We use a simple framework to organize the empirical analysis. Consider a household in location c whose income in period t depends on local rainfall R_{ct},

$$Y_{ict} = Y_{ic}\left(R_{ct}\right), \qquad \partial Y_{ic}/\partial R_{ct} > 0. \tag{1}$$

Rainfall affects income directly for farm households and indirectly for all households through local wages, prices, and demand. When rainfall falls short, the household must finance a gap between income and its consumption needs, including school fees. It can draw on savings S, insurance payouts I, transfers from relatives T, or borrowing B; any remaining gap is borne as hardship H:

$$H_{ict} = \max\left\{0,\ \bar{C}_{ic} - Y_{ic}(R_{ct}) - S_{ict} - I_{ict} - T_{ict} - B_{ict}\right\}. \tag{2}$$

This accounting delivers four predictions. *First*, unless the coping instruments fully offset the income loss, hardship rises when rainfall falls short. *Second*, if rainfall operates through the local economy rather than only through farm output, hardship rises among non-farm as well as farm households, roughly in proportion to their exposure. *Third*, because a rainfall shock is common to a community, transfers T from relatives and neighbors — who are hit by the same shock — are unlikely to rise. *Fourth*, the instruments that respond reveal which margins households actually use. If accounts, savings, and insurance are held but not drawn upon, or held too thinly to matter, households will adjust through credit, and they will turn to the lenders that are most accessible.

The framework does not determine which instruments respond; that is an empirical question. It clarifies what each finding reveals: a rise in hardship shows that coping is incomplete; the pattern across farm and non-farm households reveals the channel; and the responses of savings, insurance, transfers, and credit reveal the margins of adjustment.

V. Empirical Strategy

We estimate the following regression for respondent i in survey cluster c and wave t:

$$y_{ict} = \beta\, z_{ct} + X'_{ict}\gamma + \mu_{g(c)} + \lambda_{r(c),t} + \varepsilon_{ict}, \tag{3}$$

where y_{ict} is the outcome of interest; z_{ct} is the rainfall z-score; X_{ict} includes age, age squared, and indicators for female, married, employed, urban residence, and education; $\mu_{g(c)}$ are fixed effects for 0.5° grid cells (Tollefsen, Strand, and Buhaug, 2012); and $\lambda_{r(c),t}$ are fixed effects for $2° \times 2°$ regions (about 220 km on a side) interacted with survey year, defined within countries (1,013 region-years). The parameter of interest is β; a negative β means that the outcome rises when rainfall falls short. We report effects of a one-standard-deviation shortfall, $-\beta$, relative to the outcome mean.

The grid-cell fixed effects absorb all time-invariant differences across places, including climate, agricultural potential, and the local presence of financial institutions. The region-by-year fixed effects absorb national and regional shocks, macroeconomic conditions, national policies, changes in survey design, and regional trends in both outcomes and satellite rainfall estimates, which can drift over time. Identification therefore comes from comparing locations within the same region and year whose rainfall deviated differently from their own historical norms. Because these deviations are not chosen by households and are difficult to forecast a year ahead, they are plausibly unrelated to other determinants of hardship and borrowing conditional on the fixed effects.

The primary identification threat is that rainfall deviations are correlated with unobserved local trends in hardship or borrowing. We address this in two ways. First, we construct a placebo using rainfall in the 12 months beginning four months after fieldwork started; future rainfall cannot affect answers already given, so a significant placebo coefficient would indicate spurious correlation. Second, because the FII is a repeated cross-section, rainfall could change who is surveyed. Table 2 shows that age, sex, marital status, urban

residence, and education are unrelated to rainfall. Employment does respond to rainfall, as expected, and results are identical when we control only for predetermined characteristics.

We cluster standard errors by grid cell. Because rainfall is spatially correlated, we also report Conley (1999) standard errors with a Bartlett kernel, allowing correlation among all observations within 100, 250, or 500 km in any survey year.

Table 2. Balance: Respondent Characteristics and Rainfall

	Coefficient on rainfall z	Std. error	Mean
Age	0.0029	(0.0908)	37.363
Female	0.0046	(0.0040)	0.544
Married	0.0028	(0.0027)	0.709
Urban	−0.0026	(0.0110)	0.334
Education category	0.0144	(0.0108)	2.610
Works	−0.0114***	(0.0042)	0.510

Notes: Each row regresses the characteristic on the rainfall z-score with grid-cell and region×year fixed effects and no other controls. Standard errors clustered by grid cell. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

VI. Main Results

A. Hardship

Table 3 reports our main estimates. Columns 1–4 show that hardship rises when rainfall falls short, confirming the first prediction of the framework. A one-standard-deviation shortfall increases the probability that a child was sent home from school for unpaid fees by 1.4 percentage points, or 19 percent of the mean (column 1), and the probability of economic vulnerability by 2.7 points, or 8 percent (column 2). Going without food and going without cooking fuel each rise by about 13 percent (columns 3–4). All four estimates are significant at the 5 percent level or better with both clustered and Conley standard errors. Coping, in other words, is incomplete: rainfall shortfalls translate into lost schooling and material deprivation.

Table 3. Rainfall, Hardship, and Borrowing

	Hardship (past six months)				Borrowing (past year)	
	(1) Child sent home (fees)	(2) Economic vulnerability	(3) Went without food	(4) Went without cooking fuel	(5) Any borrowing	(6) Borrowed from NBFI
Rainfall z-score	−0.0142***	−0.0273***	−0.0205***	−0.0186***	−0.0217***	−0.0098***
	(0.0040)	(0.0078)	(0.0068)	(0.0068)	(0.0059)	(0.0026)
	[0.0048]	[0.0084]	[0.0079]	[0.0083]	[0.0071]	[0.0029]
Mean of dependent variable	0.074	0.347	0.161	0.140	0.370	0.063
Effect of a 1 SD shortfall, % of mean	+19%	+8%	+13%	+13%	+6%	+16%
Observations	157,073	235,156	157,073	157,073	370,242	363,547

Notes: All regressions include grid-cell fixed effects, $2^\circ \times 2^\circ$ region×survey-year fixed effects, and controls for age, age squared, female, married, works, urban, and education. The rainfall z-score is CHIRPS rainfall in the 12 months before fieldwork, standardized against the same window in 1981–2012 at the respondent's pixel. Hardship outcomes are observed in 2015–2016 (economic vulnerability, any of eight shortages, in 2014–2016); borrowing outcomes in all waves. NBFI: non-bank financial institution (microfinance institution, cooperative, savings and credit cooperative, post office bank). Standard errors clustered by 0.5° grid cell in parentheses; Conley (1999) standard errors with a Bartlett kernel and 250 km bandwidth, allowing correlation across all survey years, in brackets. Stars refer to clustered standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B. Borrowing

Columns 5–6 of Table 3 show that households also borrow more. A one-standard-deviation shortfall raises the probability of having borrowed in the past year by 2.2 percentage points (6 percent) and of borrowing from an NBFI by 1.0 point (16 percent). Both estimates are significant at the 1 percent level with clustered and Conley standard errors. Households thus respond to rainfall shocks on the credit margin — yet, as columns 1–4 show, hardship rises nonetheless. Because both borrowing and hardship are outcomes, we do not interpret their co-movement causally; it indicates that whatever credit households obtain in drier years is not sufficient to offset the hardship that accompanies them.

C. Effects across the Rainfall Distribution

Figure 2 replaces the z-score with indicators for seven rainfall bins, omitting normal rainfall. Borrowing is 4.7 percentage points higher in the driest bin and 4.6 points lower in the wettest, with intermediate bins approximately monotonic. Economic vulnerability is lower in all three wet bins and higher in the moderately dry bins. The relationship operates across the entire distribution: households are better off in good years and worse off in poor ones, rather than

only in extreme droughts.

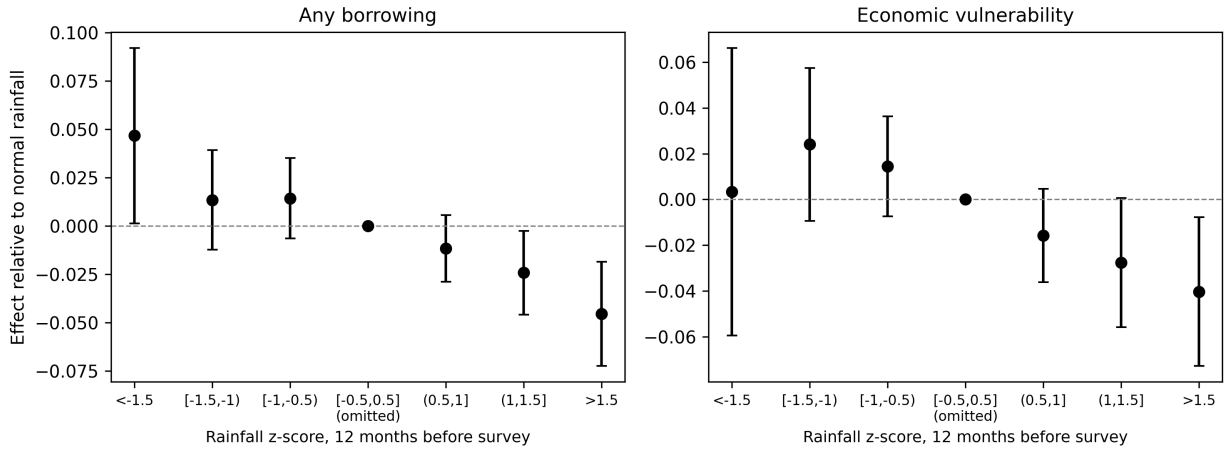


Figure 2. Borrowing and Vulnerability across the Rainfall Distribution

Notes: Coefficients and 95% confidence intervals from a single regression of each outcome on indicators for seven bins of the rainfall z-score (omitting $-0.5 \leq z \leq 0.5$), with the specification of Table 3. Standard errors clustered by grid cell.

D. Falsification and Robustness Tests

Placebo. Table 4, Panel A, reports the placebo. Rainfall in the year after fieldwork has no effect on any outcome: all six coefficients are small and statistically insignificant at the 5 percent level. When actual and future rainfall are included together, actual rainfall remains significant for every outcome while future rainfall does not. Panel B shows why the region-by-year fixed effects matter. With only country-by-year effects, future rainfall predicts children being sent home for unpaid fees and food shortages, indicating regional trends that grid-cell effects alone do not absorb when outcomes are observed in only a few waves. Region-by-year effects remove this problem. We report both specifications so that readers can see how the estimates depend on this choice.

Alternative fixed effects and samples. Table 5 shows that the results are robust to a range of alternative specifications. Replacing region-by-year with state-by-year fixed effects, which absorb state-level policies such as drought relief and the farm-loan waivers announced by several Indian states in 2017, leaves all six estimates significant. The same holds with region-by-state-by-year effects; with finer 1° region-by-year effects, which leave only 6 percent of

Table 4. Identification Checks

	Hardship (past six months)				Borrowing (past year)	
	(1) Child sent home (fees)	(2) Economic vulnerability	(3) Went without food	(4) Went without cooking fuel	(5) Any borrowing	(6) Borrowed from NBFI
Panel A. Placebo: rainfall in the 12 months starting four months after fieldwork began						
Future z-score, alone	−0.0018	−0.0117*	−0.0030	−0.0049	−0.0039	0.0009
	(0.0033)	(0.0068)	(0.0058)	(0.0062)	(0.0061)	(0.0031)
Actual z-score (joint)	−0.0143***	−0.0276***	−0.0207***	−0.0189***	−0.0151**	−0.0095***
	(0.0040)	(0.0078)	(0.0068)	(0.0068)	(0.0061)	(0.0032)
Future z-score (joint)	−0.0023	−0.0123*	−0.0038	−0.0056	−0.0034	0.0013
	(0.0033)	(0.0067)	(0.0057)	(0.0061)	(0.0060)	(0.0030)
Observations	157,073	235,156	157,073	157,073	309,520	309,520
Panel B. Country×year instead of region×year fixed effects						
Actual z-score	−0.0081***	−0.0148***	−0.0087*	−0.0180***	−0.0279***	−0.0021
	(0.0026)	(0.0053)	(0.0051)	(0.0055)	(0.0049)	(0.0016)
Future z-score (placebo)	−0.0061***	−0.0075	−0.0088**	−0.0030	−0.0026	−0.0001
	(0.0022)	(0.0046)	(0.0040)	(0.0051)	(0.0048)	(0.0020)

Notes: Panel A uses the specification of Table 3 and restricts to survey waves with rainfall observed for the placebo window (our CHIRPS extract ends in January 2019, so 2018 waves drop out). The placebo window starts four months after the fieldwork start date. Panel B replaces region×year with country×year fixed effects. Standard errors clustered by grid cell. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

the rainfall variation (except for cooking fuel); with predetermined controls only; and when dropping the 2017 waves for Kenya, Tanzania, and Uganda, which cover the 2016–2017 East African drought, or dropping Tanzania. Excluding India, the effects on school fees, economic vulnerability, any borrowing, and NBFI borrowing remain significant; within India alone, all six remain significant with state-by-year effects.

Rainfall measurement. Satellite rainfall levels can drift over time; for our Indian pixels, CHIRPS runs wetter than official India Meteorological Department figures after 2015. Region-by-year effects absorb drift that is common within a region and year, and all six results hold when rainfall is measured relative to a pixel-specific trend estimated over 1981–2012 (Table 5). The hardship outcomes, observed in 2014–2016, are less exposed to recent drift than the borrowing outcomes; in the placebo sample, which excludes 2018, the borrowing coefficient is somewhat smaller (−0.015) but remains significant. Using rainfall over the six months before fieldwork rather than 12, all outcomes respond in the same direction with smaller coefficients (school fees: −0.006, $p = 0.03$), consistent with a 12-month window

Table 5. Robustness

	Hardship (past six months)				Borrowing (past year)	
	(1) Child sent home (fees)	(2) Economic vulnerability	(3) Went without food	(4) Went without cooking fuel	(5) Any borrowing	(6) Borrowed from NBFI
Baseline (Table 3)	-0.0142^{***}	-0.0273^{***}	-0.0205^{***}	-0.0186^{***}	-0.0217^{***}	-0.0098^{***}
	(0.0040)	(0.0078)	(0.0068)	(0.0068)	(0.0059)	(0.0026)
State×year FE	-0.0104^{***}	-0.0238^{***}	-0.0156^{**}	-0.0175^{**}	-0.0169^{***}	-0.0078^{***}
	(0.0029)	(0.0082)	(0.0072)	(0.0073)	(0.0054)	(0.0018)
Region×state×year FE	-0.0082^{**}	-0.0237^{***}	-0.0201^{***}	-0.0136^{**}	-0.0155^{**}	-0.0117^{***}
	(0.0032)	(0.0084)	(0.0067)	(0.0064)	(0.0067)	(0.0029)
1° region×year FE	-0.0109^{**}	-0.0273^{***}	-0.0138^{**}	-0.0088	-0.0170^{***}	-0.0075^{**}
	(0.0046)	(0.0080)	(0.0070)	(0.0069)	(0.0061)	(0.0030)
Trend-adjusted rainfall z-score	-0.0110^{***}	-0.0258^{***}	-0.0157^{***}	-0.0140^{**}	-0.0245^{***}	-0.0091^{***}
	(0.0034)	(0.0069)	(0.0060)	(0.0057)	(0.0054)	(0.0024)
Predetermined controls only	-0.0143^{***}	-0.0275^{***}	-0.0209^{***}	-0.0188^{***}	-0.0222^{***}	-0.0099^{***}
	(0.0040)	(0.0078)	(0.0068)	(0.0068)	(0.0059)	(0.0026)
Drop Kenya, Tanzania, Uganda 2017	-0.0142^{***}	-0.0273^{***}	-0.0205^{***}	-0.0186^{***}	-0.0218^{***}	-0.0104^{***}
	(0.0040)	(0.0078)	(0.0068)	(0.0068)	(0.0061)	(0.0027)
Drop Tanzania	-0.0151^{***}	-0.0292^{***}	-0.0205^{***}	-0.0177^{**}	-0.0247^{***}	-0.0104^{***}
	(0.0041)	(0.0082)	(0.0070)	(0.0069)	(0.0061)	(0.0027)
Drop India	-0.0157^{***}	-0.0215^{**}	-0.0078	-0.0074	-0.0181^{***}	-0.0091^{**}
	(0.0051)	(0.0089)	(0.0083)	(0.0083)	(0.0063)	(0.0038)
India only, state×year FE	-0.0151^{***}	-0.0406^{***}	-0.0289^{**}	-0.0370^{***}	-0.0182^{**}	-0.0087^{***}
	(0.0047)	(0.0133)	(0.0125)	(0.0122)	(0.0076)	(0.0020)

Notes: Each cell reports the coefficient on the rainfall z-score from a separate regression; unless noted, the specification is that of Table 3. States are first-order administrative units assigned from cluster coordinates. The trend-adjusted z-score measures rainfall relative to a pixel-specific linear trend estimated over 1981–2012. Predetermined controls are age, age squared, female, urban, and education. Hardship outcomes are not observed in 2017, so dropping the 2017 East African waves does not affect them. Standard errors clustered by grid cell. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

capturing a full agricultural cycle.

Inference. Conley standard errors increase smoothly with bandwidth; at 500 km, the t-statistics are 3.3 for school fees, 3.3 for economic vulnerability, 2.3 for food, 2.2 for cooking fuel, 2.9 for any borrowing, and 3.3 for NBFI borrowing. Across 29 random splits of grid cells into halves, all 58 half-sample estimates of the borrowing effect are negative, with a 5th–95th percentile range of $[-0.032, -0.011]$.

Multiple outcomes. To address concerns about specification search, we estimate equation (3) for all 306 binary outcomes in the survey asked of at least 150,000 respondents. Rainfall yields 41 outcomes significant at 5 percent against 15 expected by chance, and 9 discoveries under the Benjamini and Hochberg (1995) false discovery rate, including four of our six main outcomes (Appendix Table A.2). A screen using placebo rainfall also yields more significant outcomes than chance, though none among our main outcomes, which is why we focus on outcomes that pass the individual placebo.

Heterogeneity by country. Appendix Table A.1 reports country-specific estimates. Children sent home for unpaid fees respond significantly in Bangladesh, Nigeria, and Pakistan and at the 10 percent level in India; borrowing responds in Bangladesh and India, and at the 10 percent level in Uganda. Tanzania, where all fieldwork dates are imputed, is the only country with a significant borrowing estimate of the opposite sign.

VII. Mechanisms

A. A Shock to the Local Economy

The second prediction of the framework concerns the channel. If rainfall affected households only through farm output, its effects would be concentrated among farm households. Table 6 shows that they are not. Effects are larger in absolute terms among households involved in agriculture, whose baseline hardship is higher, but proportional to baseline levels they are similar for the two groups: a one-standard-deviation shortfall raises the share of households with a child sent home for unpaid fees by 16 percent among farm households and 18 percent

among non-farm households, and economic vulnerability by 6 and 7 percent. Borrowing rises by similar amounts in both groups. The interaction between rainfall and farm status is insignificant for four of the six outcomes.[4]

This pattern points to rainfall as a shock to the local economy. In rural areas, poor rainfall reduces agricultural output and, with it, demand for labor, local wages, and spending at local businesses, while raising food prices (Jayachandran, 2006). Non-farm households — laborers, traders, shopkeepers — are affected through these channels, and they fall behind on school fees and face shortages much as farm households do.

Table 6. Farm and Non-Farm Households

	Hardship (past six months)				Borrowing (past year)	
	(1) Child sent home (fees)	(2) Economic vulnerability	(3) Went without food	(4) Went without cooking fuel	(5) Any borrowing	(6) Borrowed from NBFI
Farm households	−0.0164***	−0.0287***	−0.0196**	−0.0118	−0.0190***	−0.0096***
	(0.0057)	(0.0092)	(0.0086)	(0.0086)	(0.0068)	(0.0034)
Mean; effect of 1 SD shortfall, %	0.101; +16%	0.447; +6%	0.196; +10%	0.162; +7%	0.422; +5%	0.065; +15%
Non-farm households	−0.0080**	−0.0189**	−0.0150*	−0.0214**	−0.0220***	−0.0076**
	(0.0038)	(0.0096)	(0.0085)	(0.0083)	(0.0072)	(0.0032)
Mean; effect of 1 SD shortfall, %	0.044; +18%	0.256; +7%	0.122; +12%	0.115; +19%	0.318; +7%	0.060; +13%
Difference (rainfall × farm)	−0.0003	−0.0130***	−0.0005	0.0001	−0.0051*	−0.0023
	(0.0019)	(0.0037)	(0.0031)	(0.0031)	(0.0030)	(0.0019)

Notes: Coefficients on the rainfall z-score estimated separately for respondents involved and not involved in agriculture, with the specification of Table 3. The difference row is from a pooled regression including rainfall, the farm indicator, and their interaction. Standard errors clustered by grid cell. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B. Credit as the Margin of Adjustment

The third and fourth predictions concern how households cope. Table 7 shows where the additional borrowing comes from. The robust response is borrowing from NBFIs: it remains significant with state-by-year fixed effects (−0.008), while borrowing from banks, moneylenders, and other formal and informal sources does not remain significant at the 5 percent level. Within NBFIs, microfinance institutions account for a majority of the response under our main specification; cooperatives do not respond. The NBFI response is concentrated in

[4]Rainfall does not increase the share of respondents reporting any income from selling crops or livestock (−0.009, s.e. 0.005). This is an extensive-margin measure that cannot capture the value of sales and may also reflect distress sales of livestock in poor years.

Bangladesh and India (Appendix Table A.1), where microfinance is most widespread. Microfinance loans are designed for enterprise investment, so their use in poor rainfall years suggests that households draw on the most accessible formal credit to meet consumption needs.

Table 7. Sources of Borrowing

	(1) Baseline spec.	(2) State×year FE	(3) Mean	(4) 1 SD shortfall, % of mean
Any NBFI	−0.0098***	−0.0078***	0.063	+16%
	(0.0026)	(0.0018)		
Microfinance institution	−0.0058***	−0.0031**	0.032	+18%
	(0.0022)	(0.0015)		
Cooperative	−0.0003	0.0001	0.023	+1%
	(0.0013)	(0.0014)		
Bank	−0.0074***	−0.0015	0.056	+13%
	(0.0022)	(0.0018)		
Moneylender	−0.0102***	−0.0037	0.087	+12%
	(0.0037)	(0.0033)		
Any formal source	−0.0102***	−0.0011	0.117	+9%
	(0.0031)	(0.0028)		
Any informal source	−0.0139***	−0.0065*	0.215	+6%
	(0.0041)	(0.0039)		

Notes: Coefficients on the rainfall z-score. Column (1) uses the specification of Table 3; column (2) replaces region×year with state×year fixed effects. Categories overlap. Standard errors clustered by grid cell. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

C. What Does Not Respond

If accounts, savings, insurance, or remittance networks helped households absorb rainfall shocks, we would expect them to respond after shortfalls. Table 8 shows that they do not, and the estimates are precise. Using Conley standard errors and two one-sided tests, we can rule out effects of a one-standard-deviation shortfall larger than 1.5 percent of the mean in either direction on bank account ownership and about 2 percent on active bank use; increases larger than 3.8 percent in formal saving; and increases larger than 3.6 percent in insurance. Consistent with the third prediction, receiving money from relatives or friends

does not respond either (0.005, s.e. 0.006). For mobile money, whose use is low outside East Africa, the estimates are close to zero but less precise. By contrast, the same procedure gives intervals of $[+2.7\%, +9.0\%]$ for any borrowing and $[+8.5\%, +30.0\%]$ for children sent home for unpaid fees.

Taken together, the evidence shows that credit is the margin on which households adjust to rainfall shocks. Account ownership does not change, saving does not change, insurance does not change, and transfers from relatives do not rise. Households borrow, and hardship rises nonetheless.

Table 8. Accounts, Savings, and Insurance Do Not Respond

	(1) Coefficient on rainfall z	(2) Conley SE (250 km)	(3) Effect of 1 SD shortfall: 90% CI	(4) As % of mean	(5) Mean
Has bank account	−0.0001	0.0046	[−0.0076, 0.0077]	[−1.5%, +1.5%]	0.499
Active bank user	−0.0005	0.0040	[−0.0061, 0.0071]	[−1.8%, +2.1%]	0.338
Saves formally	−0.0054	0.0047	[−0.0023, 0.0132]	[−0.7%, +3.8%]	0.347
Saves informally	−0.0028	0.0029	[−0.0019, 0.0075]	[−1.9%, +7.5%]	0.100
Has insurance	0.0011	0.0037	[−0.0072, 0.0050]	[−5.1%, +3.6%]	0.140
Registered mobile money	−0.0027	0.0025	[−0.0015, 0.0068]	[−2.2%, +9.9%]	0.069
Active mobile money user	−0.0022	0.0027	[−0.0023, 0.0067]	[−3.9%, +11.3%]	0.059
Saves via mobile money	−0.0015	0.0021	[−0.0020, 0.0049]	[−5.8%, +14.2%]	0.035
For comparison:					
Any borrowing	−0.0217	0.0071	[0.0100, 0.0334]	[+2.7%, +9.0%]	0.370
Child sent home (fees)	−0.0142	0.0048	[0.0063, 0.0222]	[+8.5%, +30.0%]	0.074

Notes: Column (1) reports the coefficient on the rainfall z-score with the specification of Table 3; column (2) the Conley standard error (Bartlett kernel, 250 km). Columns (3)–(4) report the 90% confidence interval for the effect of a one-standard-deviation rainfall shortfall (the negative of the coefficient), which corresponds to two one-sided tests at the 5% level: effects outside the interval can be rejected.

D. Alternative Explanations

We consider several alternative explanations. *Changes in sample composition.* Because the FII is a repeated cross-section, rainfall could alter who is surveyed, for example through migration. Respondent characteristics are balanced with respect to rainfall (Table 2), and results are unchanged with predetermined controls only. *Regional trends.* Hardship and borrowing could trend differently across regions in ways correlated with measured rainfall. The region-by-year, state-by-year, and 1° region-by-year fixed effects absorb such trends,

and the placebo passes under our main specification. *Policy responses.* State drought relief or loan waivers could coincide with rainfall shocks; state-by-year fixed effects absorb them, and the results hold. *Measurement.* Our outcomes are extensive-margin indicators — we observe whether households borrowed, not how much or at what cost — and fieldwork dates for six country-waves are imputed, which would likely attenuate our estimates.

VIII. Magnitudes and Policy Implications

To gauge magnitudes, consider 100 households experiencing a one-standard-deviation rainfall shortfall, roughly the deviation that separates a normal year from a poor one in our data. Our estimates imply that between one and two additional households would have a child sent home from school for unpaid fees, nearly three additional households would experience at least one serious shortage, and about two additional adults would borrow. Relative to baseline rates, the school-fee effect is the largest: an increase of about one-fifth. Because school interruptions can have lasting consequences for educational attainment, these effects may extend well beyond the year of the shock.

Two policy implications follow. First, the expansion of financial access has not, by itself, equipped households to absorb covariate shocks. Accounts, savings, and insurance do not respond, while microfinance credit — designed for investment — is used to cope. Products designed for shock coping, such as index insurance (Karlan et al., 2014), emergency savings or credit lines linked to accounts, or school-fee deferral during poor seasons, may be needed to turn access into resilience. Second, because rainfall shocks reach non-farm households as much as farm households in proportional terms, responses targeted only at agricultural households will miss much of the hardship they cause.

IX. Conclusion

Rainfall shocks strike entire communities at once, limiting the informal insurance on which poor households have long relied. Over the past decade, the expansion of financial access

promised a new source of resilience. We study how households in eight developing countries cope with rainfall shocks during this expansion, using survey data on 370,242 adults linked to satellite rainfall.

We find that rainfall shortfalls impose real hardship. A one-standard-deviation shortfall raises the probability that a child is sent home from school for unpaid fees by 19 percent and economic vulnerability by 8 percent. These effects are proportionally similar for farm and non-farm households, consistent with a shock that propagates through the local economy rather than only through farm output. Placebo tests using future rainfall, balance tests, and a wide range of fixed effects and inference methods support a causal interpretation.

Households respond to these shocks by borrowing, especially from microfinance institutions in South Asia. By contrast, bank accounts, savings, insurance, and transfers from relatives do not respond, and we can rule out economically meaningful effects on accounts and formal saving. Credit is the margin of adjustment, and it is not enough to prevent hardship.

These findings suggest that financial access and financial resilience are not the same. Turning the former into the latter will likely require instruments designed for the shocks that poor households face most often — shocks that, as our results show, reach well beyond the farm.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with writing and debugging data-processing and statistical code, organizing the literature, and drafting and editing the manuscript text. After using this tool, the authors reviewed, verified, and edited all content, including code, estimates, and references, and take full responsibility for the content of the published article.

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Appendix

Table A.1. Results by Country

	(1) Child sent home	(2) Econ. vulnerability	(3) Any borrowing	(4) NBFI
Bangladesh	-0.0151^{***}	-0.0324	-0.0500^{***}	-0.0267^{***}
	(0.0055)	(0.0249)	(0.0126)	(0.0091)
India	-0.0119^{*}	-0.0392^{***}	-0.0262^{**}	-0.0104^{***}
	(0.0063)	(0.0149)	(0.0102)	(0.0035)
Indonesia	0.0000	-0.0379^{*}	-0.0279	0.0028
	(0.0033)	(0.0198)	(0.0171)	(0.0090)
Kenya	0.0005	-0.0260	-0.0181	-0.0098
	(0.0253)	(0.0235)	(0.0193)	(0.0088)
Nigeria	-0.0720^{***}	-0.0681^{***}	-0.0046	-0.0005
	(0.0260)	(0.0258)	(0.0157)	(0.0036)
Pakistan	-0.0093^{***}	-0.0030	0.0056	-0.0044^{*}
	(0.0033)	(0.0193)	(0.0109)	(0.0025)
Tanzania	-0.0009	0.0120	0.0448^{***}	0.0034
	(0.0189)	(0.0243)	(0.0120)	(0.0058)
Uganda	-0.0281	0.0066	-0.0255^{*}	0.0018
	(0.0232)	(0.0164)	(0.0134)	(0.0062)

Notes: Coefficients on the rainfall z-score from country-specific regressions with the specification of Table 3. Fieldwork start dates for Tanzania are imputed. Standard errors clustered by grid cell; p-values use the t distribution with $G-1$ degrees of freedom. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.2. Screening All Outcomes

	Outcomes tested	$p < 0.05$	Expected by chance	BH $q < 0.05$	BH $q < 0.10$
Rainfall z-score	306	41	15	9	15
Placebo: future rainfall	306	37	15	3	7
Permuted rainfall	306	18	15	0	0

Notes: Each of 306 binary outcomes asked of at least 150,000 respondents is regressed on the shock with the specification of Table 3. BH: Benjamini–Hochberg (1995) false discovery rate. The placebo uses rainfall in the 12 months starting four months after fieldwork began (sample restricted to waves where it is observed); the permuted shock reassigns z-scores across survey clusters within grid cells. Of the nine rainfall discoveries at $q < 0.05$, one (lack of fertilizer) fails the placebo; the three placebo discoveries are saving in cash or property, two reasons for using mobile money, and sending picture messages. Full results in the Online Appendix.

Table A.3. Survey Waves and Fieldwork Start Dates

Country	Survey year	Fieldwork start used	Source
Bangladesh	2014	01 Sep 2014	FII report
Bangladesh	2015	01 Sep 2015	FII report
Bangladesh	2016	01 Aug 2016	FII report
Bangladesh	2017	29 Jul 2017	FII report
Bangladesh	2018	01 Sep 2018	Imputed (1 Sep)
India	2014	14 Sep 2014	FII report
India	2015	01 Jun 2015	FII report
India	2016	05 Sep 2016	FII report
India	2017	16 Aug 2017	FII report
India	2018	01 Sep 2018	FII report
Indonesia	2014	01 Aug 2014	FII report
Indonesia	2015	01 Aug 2015	FII report
Indonesia	2016	01 Sep 2016	Imputed (1 Sep)
Indonesia	2018	01 Dec 2018	FII report
Kenya	2014	01 Sep 2014	FII report
Kenya	2015	01 Sep 2015	FII report
Kenya	2016	01 Aug 2016	FII report
Kenya	2017	16 Jun 2017	FII report
Nigeria	2014	01 Jul 2014	FII report
Nigeria	2015	01 Aug 2015	FII report
Nigeria	2016	01 Aug 2016	FII report
Nigeria	2017	01 Aug 2017	FII report
Pakistan	2014	01 Sep 2014	FII report
Pakistan	2015	01 Sep 2015	FII report
Pakistan	2016	01 Sep 2016	FII report
Pakistan	2017	03 Sep 2017	FII report
Tanzania	2014	01 Sep 2014	Imputed (1 Sep)
Tanzania	2015	01 Sep 2015	Imputed (1 Sep)
Tanzania	2016	01 Sep 2016	Imputed (1 Sep)
Tanzania	2017	01 Sep 2017	Imputed (1 Sep)
Uganda	2014	01 Jun 2014	FII report
Uganda	2015	01 Jul 2015	FII report
Uganda	2016	01 Jul 2016	FII report
Uganda	2017	01 Jul 2017	FII report

Notes: Fieldwork start months are taken from InterMedia FII annual and technical reports. Where a report gives only a month, the first day of that month is used. For the six country-waves whose start dates could not be located in the reports, we impute 1 September, the modal start month across documented waves. Rainfall shocks are measured over the 365 days before the fieldwork start date. Results restricted to confirmed dates are reported in Table 5.